

YEAR 7 MATHEMATICS

Chapter 5B

Equations

MIXED PRACTICE WORKSHEETS

5B

How to use this booklet

- 1 Each lesson is **one double-sided sheet**. Write your name, class and date at the top.
- 2 Then work down the **columns**, left to right. They get harder as you go: **Easy** **Easy** **Medium** on page 1, then **Medium** **Challenging** **Challenging** on page 2.
- 3 Most questions practise **today's skill**. A question marked **from 1.03** comes from an earlier lesson, to keep it fresh. Stuck? Look back at that lesson.
- 4 Each question tells you what to do: read it carefully. Show your working on the lines. The first question with working in each Easy column shows you how to set it out.

WE DO

The grey box at the top. Your teacher shows the key idea with two or three examples. **Copy every step** into the space.

Contents and progress tracker

Lesson	Topic	Page	Date	Done
5.10	Guess, check and improve	3		☐
5.11	Solving one-step equations	5		☐
5.12	Solving two-step equations	7		☐
5.13	Equations and problem solving	9		☐

In this chapter you will

- check whether a value is a solution
- solve equations by guess, check and improve
- solve one-step and two-step equations using inverse operations
- write and solve equations to solve problems

About the artwork

The pictures in this booklet are parts of the **Mandelbrot set**, drawn by a computer that repeats one simple rule, $z \rightarrow z^2 + c$, thousands of times for every point. Every booklet shows a different part of the set: this one is the valley between the main body and the left bulb, where the seahorses start. Maths can make beautiful things.

Acknowledgement of Country

We acknowledge the Bundjalung people, the Traditional Custodians of the land on which Kingscliff High School stands, and pay our respects to Elders past and present.

NAME _____ CLASS _____ DATE _____

WE DO

Watch your teacher.
Copy every step.

KEY IDEA

An equation is true for one value of the pronumeral: the solution. Guess, check and improve: try a value, substitute it, see if it is too big or too small, then try again.

E1 Solve in your head:
 $x + 7 = 12$

E2 Is $m = 4$ a solution of
 $3m + 2 = 15$?

E3 Guess, check and improve: $6p + 5 = 77$

Easy

Questions 1–9

1 Solve in your head: $2y = 6$

2 Solve in your head: $\frac{x}{5} = 11$

3 Is $x = 8$ a solution of
 $3x + 1 = 25$?

4 Solve in your head: $w + 2 = 11$

5 Solve in your head: $\frac{x}{4} = 12$

6 Is $n = 6$ a solution of
 $5n + 8 = 43$?

7 Solve in your head: $10b = 70$

8 Solve in your head: $\frac{n}{2} = 2$

9 Find k^2 when $k = 3$.

from 5.05

Easy

Questions 10–18

10 Solve in your head: $\frac{n}{6} = 3$

11 Is $p = 3$ a solution of
 $2p + 3 = 9$?

12 Solve in your head: $10w = 10$

13 Write 35% as a decimal.

from 4.10

14 Is $y = 8$ a solution of
 $2y + 1 = 17$? Show the check.

$2 \times 8 + 1 =$ _____

yes / no: _____

15 Solve in your head: $\frac{t}{6} = 12$

16 Is $p = 6$ a solution of
 $3p + 9 = 27$?

17 $bn \div b$

from 5.09

18 Solve in your head: $4p = 12$

Medium

Questions 19–25

19 Is $y = 8$ a solution of
 $3y + 7 = 28$? Show the check.

20 Solve $6y + 16 = 94$. Your first guess is 11. Is it too big or too small?

21 $49 + 42 + 51 + 16$

from 5.01

22 Solve by guess and check:
 $x^2 = 144$, where x is positive.

23 Is $w = 2$ a solution of
 $2w + 5 = 11$? Show the check.

24 A rectangle is $7b$ cm long and
 $5n$ cm wide. Write its area.

from 5.08

25 Solve $7m + 10 = 101$. Your first guess is 10. Is it too big or too small?

Medium

Questions 26–35

- 26 Solve $6a + 12 = 48$. Your first guess is 5. Is it too big or too small?

- 27 Solve by guess and check: $y^2 = 25$, where y is positive.

- 28 Write $4\frac{2}{5}$ as an improper fraction.

from 4.01

- 29 Is $b = 6$ a solution of $6b + 5 = 35$? Show the check.

- 30 Work out $5 \times (30 + 6)$ two ways.

from 5.02

- 31 Guess, check and improve: $8x + 8 = 112$. Record each guess.

- 32 Solve $3n + 12 = 42$. Your first guess is 7. Is it too big or too small?

- 33 Simplify $12x - 5x$.

from 5.07

- 34 Solve by guess and check: $b^2 = 49$, where b is positive.

- 35 Find $4n + 9$ when $n = -3$.

from 5.05

Challenging

Questions 36–44

- 36 Guess, check and improve: $7x + 14 = 56$. Record each guess.

- 37 Two numbers that are next to each other add to 17. Find them by guess, check and improve.

- 38 Jack scored 50% on a test out of 180. How many marks is that?

from 4.11

- 39 A number multiplied by itself, plus the number, is 56. Find the number.

- 40 $9a \div a$

from 5.09

- 41 Guess, check and improve: $4x + 6 = 42$. Record each guess.

- 42 Two numbers that are next to each other add to 25. Find them by guess, check and improve.

- 43 Write $7mn$ in expanded form.

from 5.03

- 44 A number multiplied by itself, plus the number, is 90. Find the number.

Challenging

Questions 45–54

- 45 Two numbers that are next to each other add to 21. Find them by guess, check and improve.

- 46 A number multiplied by itself, plus the number, is 20. Find the number.

- 47 $3 \times \frac{2}{3}$

from 4.06

- 48 Guess, check and improve: $3a + 18 = 57$. Record each guess.

- 49 Write an expression: 4 less than c

from 5.04

- 50 Solve by guess and check: $n^2 = 144$, where n is positive.

- 51 Two numbers that are next to each other add to 15. Find them by guess, check and improve.

- 52 Are $6a^2$ and $6a$ like terms?

from 5.07

- 53 A number multiplied by itself, plus the number, is 12. Find the number.

- 54 Find ab when $a = 9$ and $b = 6$.

from 5.05

5.11 Solving one-step equations

YEAR 7 · CHAPTER 5B: EQUATIONS · PAGE 1 OF 2

NAME _____ CLASS _____ DATE _____

WE DO

KEY IDEA

To solve an equation, do the opposite operation to both sides: + undoes −, − undoes +, ÷ undoes ×, × undoes ÷. Check by substituting the answer.

Watch your teacher.
Copy every step.

E1 Solve $x + 9 = 23$.

E2 Solve $4m = 36$.

E3 Solve $\frac{p}{5} = 7$.

Easy

Questions 1–9

1 Solve $t + 2 = 21$.

2 Solve $n - 20 = 11$.

3 What undoes $\div 5$?

4 Solve $k + 8 = 30$.

5 Solve $n - 15 = 9$.

6 What undoes $+ 7$?

7 Solve $y + 2 = 23$.

8 Solve $x - 17 = 15$.

9 Solve in your head: $\frac{t}{5} = 6$

from 5.10

Easy

Questions 10–18

10 Solve $p - 20 = 29$.

11 What undoes $- 4$?

12 Solve $a + 5 = 17$.

13 Evaluate 7^2 .

from 3.07

14 Solve $3m = 12$.

$3m \div 3 = \underline{\hspace{2cm}}$

$m = \underline{\hspace{2cm}}$

15 Solve $a - 1 = 17$.

16 What undoes $\times 3$?

17 $kx \div k$

from 5.09

18 Solve $t + 18 = 24$.

Medium

Questions 19–26

19 Solve $4m = 36$.

20 Solve $\frac{x}{8} = 8$.

21 $35 + 47 + 65 + 40$

from 5.01

22 Solve $y + 4 = 3$.

23 Solve $8a = 24$.

24 A rectangle is $5b$ cm long and $3n$ cm wide. Write its area.

from 5.08

25 Solve $\frac{x}{4} = 12$.

26 Solve $w + 12 = 8$.

Medium

Questions 27–37

27 Solve $\frac{w}{6} = 2$.

28 Solve $t + 10 = 5$.

29 $\frac{6}{9} - \frac{1}{9}$

from 4.03

30 Solve $8k = 24$.

31 Work out $5 \times (50 + 3)$ two ways.

from 5.02

32 Solve $2m = 23$. Give the answer as a fraction or mixed numeral.

33 Solve $\frac{a}{5} = 8$.

34 Is $w = 3$ a solution of $3w + 6 = 12$?

from 5.10

35 Solve $n + 5 = 0$.

36 Simplify $\frac{16ay}{4y}$.

from 5.09

37 Solve $7k = 77$.

Challenging

Questions 38–47

38 Solve $9p = 25$. Give the answer as a fraction or mixed numeral.

39 Solve $3a = -9$.

40 202×6

from 3.02

41 Solve $56 = k + 47$.

42 $4c \times c$

from 5.08

43 Ruby solved $6a = 12$ and got $a = 6$. What went wrong? What is the answer?

44 Solve $2n = 29$. Give the answer as a fraction or mixed numeral.

45 Write $9nx$ in expanded form.

from 5.03

46 Solve $8w = -32$.

47 Is $m = 9$ a solution of $4m + 10 = 46$? Show the check.

from 5.10

Challenging

Questions 48–57

48 Solve $5t = -15$.

49 Solve $34 = t + 26$.

50 $50\,000 \div 1\,000$

from 3.03

51 Lena solved $2t = 6$ and got $t = 4$. What went wrong? What is the answer?

52 Write an expression: 3 less than c

from 5.04

53 Solve $4y = 11$. Give the answer as a fraction or mixed numeral.

54 Solve $w + 11 = 4$.

55 $16m \div 4$

from 5.09

56 Solve $8p = -40$.

57 $5mx \times 3m$

from 5.08

5.12 Solving two-step equations

YEAR 7 · CHAPTER 5B: EQUATIONS · PAGE 1 OF 2

NAME _____ CLASS _____ DATE _____

WE DO

KEY IDEA

A two-step equation: undo the + or – first, then the $\times$ or $\div$. Do the same to both sides each time, then check your answer.

Watch your teacher.
Copy every step.

E1 Solve $3x + 5 = 26$.

E2 Solve $5m - 4 = 31$.

E3 Solve $2(a + 3) = 18$.

Easy

Questions 1–9

1 What do you undo first in $5w + 9 = 34$?

2 Solve $3m + 7 = 16$.

3 Check: is $p = 7$ the solution of $5p - 10 = 20$?

4 What do you undo first in $8a - 3 = 27$?

5 Solve $5y + 7 = 32$.

6 Check: is $y = 9$ the solution of $6y - 6 = 42$?

7 What do you undo first in $4y - 8 = 31$?

8 Solve $5y + 3 = 8$.

9 Solve $p - 10 = 17$.

from 5.11

Easy

Questions 10–18

10 Solve $5a + 7 = 17$.

11 Check: is $t = 5$ the solution of $6t - 12 = 12$?

12 What do you undo first in $6p + 13 = 58$?

13 Write 5 out of 18 as a fraction in simplest form.

from 4.12

14 Solve $9m - 20 = 43$.

$9m = \underline{\hspace{2cm}}$

$m = \underline{\hspace{1cm}} \div \underline{\hspace{1cm}} = \underline{\hspace{1cm}}$

15 Solve $2a + 5 = 11$.

16 Check: is $b = 2$ the solution of $2b - 5 = -1$?

17 Solve in your head: $\frac{x}{4} = 4$

from 5.10

18 What do you undo first in $4m + 9 = 42$?

Medium

Questions 19–26

19 Solve $8x + 15 = 39$.

20 Solve $\frac{t}{2} + 5 = 11$.

21 $18 + 83 + 82 + 23$

from 5.01

22 Solve $2(n + 8) = 20$.

23 Solve $5y + 16 = 46$.

24 Simplify $\frac{20n}{5}$.

from 5.09

25 Solve $\frac{m}{4} + 6 = 10$.

26 Solve $4(p + 3) = 48$.

Medium

Questions 27–37

27 Solve $\frac{a}{2} + 3 = 6$.

28 Solve $6(y + 7) = 96$.

29 $7 \times \frac{1}{9}$

from 4.06

30 Solve $7m - 18 = 59$.

31 Work out $9 \times (20 + 8)$ two ways.

from 5.02

32 Solve $3a + 10 = 1$.

33 Solve $\frac{m}{6} + 5 = 13$.

34 What undoes -4 ?

from 5.11

35 Solve $4(n + 8) = 60$.

36 Solve by guess and check:
 $a^2 = 49$, where a is positive.

from 5.10

37 Solve $8n - 14 = 18$.

Challenging

Questions 38–47

38 Solve $6x + 7 = -29$.

39 Solve $9x + 12 = 24$. Give the answer as a mixed numeral.

40 412×4

from 3.02

41 Solve $2(t - 4) = 6$.

42 $4b \div b$

from 5.09

43 Mia solved $6t + 5 = 23$ and got $t = 12$. What went wrong?

44 Solve $6b + 4 = -50$.

45 Write $3cx$ in expanded form.

from 5.03

46 Solve $8b + 11 = 25$. Give the answer as a mixed numeral.

47 Solve $3a = 33$.

from 5.11

Challenging

Questions 48–57

48 Solve $3m + 15 = 22$. Give the answer as a mixed numeral.

49 Solve $3(x - 7) = -15$.

50 What is 10% of 358?

from 4.11

51 Priya solved $4p + 7 = 27$ and got $p = 16$. What went wrong?

52 Write an expression: 10 less than c

from 5.04

53 Solve $3y + 11 = 5$.

54 Solve $5(k + 9) = 105$.

55 Solve in your head: $y - 4 = 5$

from 5.10

56 Solve $8t + 5 = 49$. Give the answer as a mixed numeral.

57 $54ct \div 6t$

from 5.09

NAME _____ CLASS _____ DATE _____

WE DO

KEY IDEA

To solve a problem with an equation: choose a pronumeral for the unknown, write an equation from the words, solve it, then answer in a sentence.

Watch your teacher.
Copy every step.

E1 A number times 4, plus 7, is 43. Find the number.

E2 6 equal packs hold 72 pencils. How many pencils are in each pack?

E3 Two angles of a triangle are 50° and 65° . Find the third angle.

Easy

Questions 1–9

1 Write an equation: A number plus 4 is 19.

2 A number plus 2 is 14. What is the number?

3 Write an equation: 5 times a number is 20.

4 7 times a number is 77. What is the number?

5 Write an equation: A number minus 3 is 13.

6 A number minus 4 is 5. What is the number?

7 Write an equation: A number divided by 8 is 7.

8 A number plus 4 is 21. What is the number?

9 Solve $2k + 4 = 12$.

from 5.12

Easy

Questions 10–18

10 4 times a number is 68. What is the number?

11 Write an equation: A number plus 4 is 12.

12 Double a number, plus 6, is 44. Write an equation and find the number.

$$______ n + ______ = ______$$

$$n = ______$$

13 Write $3\frac{1}{2}$ as an improper fraction.

from 4.01

14 A number minus 3 is 12. What is the number?

15 Write an equation: 7 times a number is 112.

16 Three times a number, minus 4, is 50. Write an equation and find the number.

17 Write an expression: 7 less than b

from 5.04

18 A number plus 2 is 22. What is the number?

Medium

Questions 19–24

19 A number is multiplied by 5, then 9 is added. The answer is 59. Write an equation and find the number.

20 7 equal packs hold 203 pencils. How many pencils are in each pack?

21 $22 + 46 + 78 + 11$

from 5.01

22 Two angles of a triangle are 25° and 51° . Write an equation and find the third angle.

23 Double a number, plus 3, is 37. Write an equation and find the number.

24 Solve $\frac{k}{8} = 7$.

from 5.11

Medium

Questions 25–32

- 25 7 equal packs hold 56 pencils. How many pencils are in each pack?

- 26 Two angles of a triangle are 68° and 70° . Write an equation and find the third angle.

- 27 A class has 12 girls and 15 boys. What fraction of the class are girls?

from 4.08

- 28 Three times a number, minus 5, is 31. Write an equation and find the number.

- 29 Work out $5 \times (50 + 3)$ two ways.

from 5.02

- 30 A plumber charges \$25 to call, plus \$25 an hour. The bill is \$125. How many hours did the plumber work?

- 31 3 equal packs hold 87 pencils. How many pencils are in each pack?

- 32 Solve in your head: $\frac{t}{2} = 9$

from 5.10

Challenging

Questions 33–40

- 33 A plumber charges \$30 to call, plus \$49 an hour. The bill is \$275. How many hours did the plumber work?

- 34 A rectangle is 2 cm longer than it is wide. Its perimeter is 64 cm. Find its width.

- 35 Write 0.8 as a percentage and as a fraction.

from 4.10

- 36 Aisha is 6 years older than Sam. Their ages add to 18. How old is Sam?

- 37 Write an expression: 6 times a

from 5.04

- 38 Three numbers in a row add to 39. Find them.

- 39 A plumber charges \$25 to call, plus \$25 an hour. The bill is \$75. How many hours did the plumber work?

- 40 Write $4np$ in expanded form.

from 5.03

Challenging

Questions 41–49

- 41 A rectangle is 8 cm longer than it is wide. Its perimeter is 56 cm. Find its width.

- 42 Ali is 6 years older than Sam. Their ages add to 20. How old is Sam?

- 43 Write 35 in Roman numerals.

from 3.06

- 44 Three numbers in a row add to 36. Find them.

- 45 Find m^2 when $m = 5$.

from 5.05

- 46 A plumber charges \$40 to call, plus \$48 an hour. The bill is \$136. How many hours did the plumber work?

- 47 Two angles of a triangle are 47° and 20° . Write an equation and find the third angle.

- 48 Is $b = 7$ a solution of $5b + 10 = 50$?

from 5.10

- 49 A rectangle is 7 cm longer than it is wide. Its perimeter is 54 cm. Find its width.